



Rapid high mass resolution mass spectrometry using matrix-assisted ionization



Sarah Trimpin^{a,b,*}, Shameemah Thawoos^a, Casey D. Foley^a, Daniel W. Woodall^a, Jing Li^a, Ellen D. Inutan^a, Paul M. Stemmer^c

^a Department of Chemistry, Wayne State University, Detroit, MI 48202, USA

^b Cardiovascular Research Institute, Wayne State University School of Medicine, Detroit, MI 48202, USA

^c Proteomics Core Facility, Institute of Environmental Health Sciences, Wayne State University, Detroit, MI 48202, USA

ARTICLE INFO

Article history:

Received 8 December 2015

Accepted 29 January 2016

Available online 4 February 2016

Keywords:

Matrix-assisted

Laserspray

Ionization

Protein

Mass spectrometry

High resolution

ABSTRACT

Matrix-assisted ionization (MAI) is demonstrated to be a robust and sensitive analytical method capable of analyzing proteins such as cholera toxin B-subunit and pertussis toxin mutant from conditions containing relatively high amounts of inorganic salts, buffers, and preservatives without the need for prior sample clean-up or concentration. By circumventing some of the sample preparation steps, MAI simplifies and accelerates the analytical workflow for biological samples in complex media. The benefits of multiply charged ions characteristic of electrospray ionization (ESI) and the robustness of matrix-assisted laser desorption/ionization (MALDI) can be obtained from a single method, making it well suited for analysis of proteins and other biomolecules at ultra-high resolution as demonstrated on an Orbitrap Fusion where protein subunits were resolved for which MALDI-time-of-flight failed. MAI results are compared with those obtained with ESI, MALDI, and laserspray ionization methods and fundamental commonalities discussed.

© 2016 Elsevier Inc. All rights reserved.

1. Introduction

Mass spectrometry (MS) has been established as a prominent analytical method for the study of biomolecules, particularly in the analysis of proteins [1]. Analysis of membrane-associated proteins is of significant interest because of their role in the function of many drugs and biomolecules [2]. Matrix-assisted laser desorption/ionization (MALDI) and electrospray ionization (ESI) are soft ionization methods widely used in the analysis of biomolecules because of their ability to accurately identify analytes from a variety of media. ESI generates multiply charged analyte ions from solution by applying a high voltage, whereas MALDI generates predominantly singly charged molecules from solid-phase matrix:analyte crystals with the use of a laser. Matrix-assisted ionization (MAI) is a recently introduced [3] ionization method that generates ESI-like multiply charged ions, but from solid-phase matrix:analyte crystals similar to MALDI. Contrary to ESI and MALDI, MAI does not require the use of high voltage or a laser to generate ions, using only the inherent vacuum [4] of the mass spectrometer and a

MAI matrix [5]. The production of multiply charged analyte ions shifts the observed signals to a lower mass-to-charge ratio (m/z), allowing MAI to be used with atmospheric pressure (AP) high-performance mass spectrometers that have a limited mass range but improved mass resolution [6–9]. MAI is operational from both AP and intermediate pressure with either minor or no instrument modifications aside from removing the AP source housing and overriding interlocks [10]. Laserspray ionization (LSI), is a special case of MAI, in which a laser is used to ablate matrix:analyte crystals into the AP inlet of the mass spectrometer where ionization is initiated [11–14].

Biological samples commonly contain high concentrations of inorganic salts and buffers to enhance solubility and stabilize pH. These constituents suppress ionization in MS partially due to charge competition, salt adduction, and reduced analyte desolvation/declustering [15], most noticeably with ESI. For this reason sample purification, desalting, and concentration methods such as reverse-phase liquid chromatography (LC) [16], solid phase extraction [17], or chip-based microfluidic devices [18] are often used to reduce the sample complexity prior to ionization. Commonly used buffers in biochemistry such as TRIS and phosphate salt buffers are not compatible for use with ESI, and high salt content can cause clogging of the electrospray capillary [19]. As a result, a buffer exchange step is often required prior to analysis.

* Corresponding author at: Department of Chemistry, Wayne State University, Detroit, MI 48202, USA.

E-mail address: stimpin@chem.wayne.edu (S. Trimpin).

While these methods are effective, they are time consuming and usually result in loss of analyte during sample preparation.

Predominantly multiply charged protonated analyte ions are observed using MAI, even in the presence of inorganic salts and sample preservatives such as ethylenediaminetetraacetic acid (EDTA). MAI is demonstrated here for proteins, including membrane-associated bacterial toxins, for which a direct and robust method capable of ionizing proteins from salty or buffered solutions without sample preparation or cleanup is desirable.

2. Experimental

2.1. Mass spectrometers

Analyses were performed on Waters SYNAPT G2 and SYNAPT G2S mass spectrometers (Manchester, UK) with the ESI housing removed and interlocks overridden as previously described [10]. The temperature of the Waters Z-Spray source block was typically set between 30 and 50 °C to achieve maximum sensitivity [20]. Thermo LTQ Velos and Orbitrap Fusion (San Jose, CA) mass spectrometers were also used but typically operated at 150 °C and 100 °C, respectively. ESI was performed at a flow rate of 10 $\mu\text{L min}^{-1}$ and at a capillary voltage of 3.0 kV. MALDI analysis were performed with sinapinic acid as the matrix on a Bruker MALDI-TOF Ultraflexxtreme (Bremen, Germany).

2.2. Materials and sample preparation

Cholera toxin B-subunit (11,605 Da), myoglobin (16,951 Da), cytochrome C (12,359 Da), lysozyme (14,305 Da), ubiquitin (8,565 Da), bovine insulin (5,734 Da), 2-nitrobenzonitrile (2-NBN), 2-nitrophenylglycine (2-NPG), and sinapinic acid were purchased from Sigma Aldrich (St. Louis, MO). Pertussis toxin mutant, consisting of 6 subunits (S1 26,017 Da; S2 21,839 Da; S3 21,751 Da; S4 12,061 Da; S5 11,747 Da) was purchased from List Biological Laboratories (Campbell, CA). Calmodulin (16706 Da) was provided by the Proteomics Core Facility of Wayne State University (Detroit, MI) using a protocol which involved extraction from *Escherichia coli*, and storage in a 20 mM Tris buffer at pH 7.5 and 2.0 mM ethylene glycol tetraacetic acid at pH 7.5. It was used without purification. LC–MS grade water, HPLC grade methanol (MeOH), HPLC grade acetonitrile (ACN), and glacial acetic acid were purchased from Fisher Scientific (Fairlawn, NJ). 3-Nitrobenzonitrile (3-NBN) was purchased from TCI America (Portland, OR). Myoglobin and cytochrome C were prepared by reconstituting the lyophilized powder in water then further diluted in 50% aqueous MeOH with 1% acetic acid to concentrations ranging from 50 nM to 1 μM and used without further purification. Lysozyme, ubiquitin, and bovine insulin were prepared in 50% aqueous ACN with 1% acetic acid. Calmodulin extract was diluted in 50% aqueous MeOH with 1% acetic acid. 3-NBN and 2-NBN matrix solutions were both prepared to a concentration of 100 mg/mL in ACN. Prior to analysis, matrix and analyte solutions were mixed 1:1 (v:v) in a microcentrifuge tube and vortexed briefly (ca. 3 s). After mixing, 1 or 1.5 μL of the matrix:analyte solution was drawn into the end of a disposable pipet tip, and allowed to crystallize for ca. 10 s on the end of the pipet tip upon which crystals were subsequently drawn into the aperture of the mass spectrometer by the pressure differential between atmosphere and the vacuum of the mass spectrometer.

A solution was prepared from an allergy pill (5 ppm) containing fexofenadine (MW 501.3 Da) by dissolving the tablet in 50% aqueous methanol. This solution was used as a standard and mixed with 75% aqueous acetonitrile. Blood was extracted from a blood spot on a Kimwipe using a solution of 100 mg/50 μL 3-NBN and 1 μL of this solution was introduced to the mass spectrometer

using a pipette tip. Undiluted urine was mixed with 100 mg/50 μL 3-NBN solution in 1:1 (v:v) and 1 μL of the solution was used as above.

A sample containing 40 μM cholera toxin B-subunit in 50 mM TRIS buffer (pH 7.5), 200 mM NaCl, 1 mM EDTA and 3 mM NaN_3 was diluted 40-fold to a final protein concentration of 1 μM (1 pmol/ μL) and analyzed by MAI-MS. Pertussis toxin mutant was analyzed by MAI-MS from a sample containing 0.5 mg/mL of the protein in 10 mM sodium phosphate buffer (pH 7.0) with 50 mM NaCl. The final concentration of the protein after dilution was 1 μM . The mutations are present on the S1 subunit (R9K and E129A) to inactivate the enzymatic activity, making it safer to handle [21]. Further experimental details are provided in the Supplementary Information.

3. Results and discussion

A direct comparison between MAI, without and with the use of a laser, is shown for lysozyme using different MAI matrices, previously described [4,22], and inlet tube temperatures of 150 °C (Fig. 1.A) and 250 °C (Fig. 1.B and Fig. S1). In MAI, the matrix:analyte crystals are physically introduced into the AP inlet without the use of a laser, whereas the use of a laser to ablate the sample into the inlet has been referred to as LSI and is shown in Scheme 1 and Movie S1. A MAI matrix is crystallized together with the analyte on a glass plate, similar to MALDI sample preparation, and the glass plate is adhered to the AP inlet of an LTQ Velos mass spectrometer by the pressure differential so that the matrix:analyte crystals experience approximately the same sub-AP for MAI and LSI acquired using the same sample (Fig. 1). At an inlet tube temperature of 150 °C using the 2-NBN matrix (Fig. 1.A) the MAI total ion signal abundance is ca. 1/3 of the intensity of the LSI total ion signal abundance whereas at an inlet tube temperature of 250 °C the total ion signal abundance ratio of MAI and LSI is ca. 8/1 (Fig. 1.B) as shown in the total ion chromatograms (TICs) (Fig. 1.A.(i) and 1.B.(i)). As visualized in the movie provided in the Supplementary file, MAI occurs both when the glass plate is placed on the inlet tube connector and when the laser is fired, as observed by the first

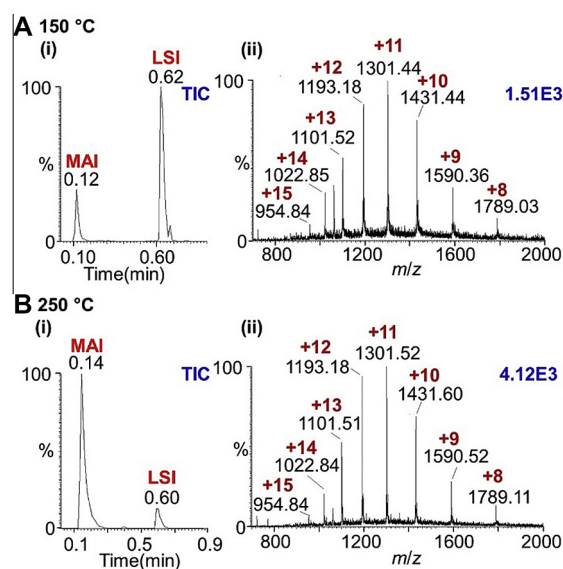
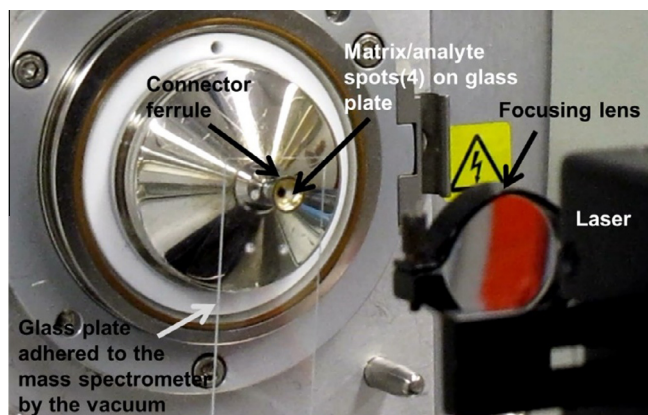


Fig. 1. LSI and MAI-MS consuming 5 pmol of lysozyme (Theoretical: 14305 Da/ Experimental: 14305 Da) prepared in water with 0.1% formic acid and matrix 2-NBN using a Thermo LTQ Velos mass spectrometer with inlet temperature set at (A) 150 °C and (B) 250 °C. (i) TIC of acquiring first MAI followed by LSI; (ii) LSI mass spectra summed over the respective area under the TIC.



Scheme 1. Operation of LSI and MAI on a Thermo LTQ Velos mass spectrometer with ionization depending on the use of a matrix and temperature/pressure. A movie is included in the Supplementary information for the special case of LSI and showing performance characteristics using the same matrix (3-NBN)/analyte spot with (LSI) and without (MAI) the use of a laser.

and second signal in the TIC, respectively. The relative ion intensities give an indication of the effectiveness of MAI *without* employing the laser, and especially with increasing the inlet tube temperature using the 2-NBN matrix. Performing the same experiment (Scheme 1), but with the matrix 2-NPG (a useful LSI matrix, even operating from high vacuum [22]), the sample sublimed rapidly without the use of a laser when exposed to vacuum condition at an inlet tube temperature of 150 °C (Fig. S1); a laser was not needed at either 150 or 250 °C. In all cases multiply charged ions are formed, keeping m/z values well within the mass range of the mass spectrometer.

A direct comparison of MAI and LSI on the SYNAPT G2S shows increased ubiquitin ion abundance using MAI (5.12E3) versus LSI (7.53E2) (Fig. S2). We attribute this difference to a more efficient introduction of the matrix:analyte into the mass spectrometer inlet using MAI. In LSI, the laser beam can be focused to ablate only small areas providing high spatial resolution. The inclusion of ion mobility spectrometry (IMS) notably increases the signal-to-noise ratio (Fig. S2.A.(ii) and S2.B.(ii)).

Analyses from buffered protein solutions are applicable as is shown for positive mode measurements of calmodulin from an *Escherichia coli* extract with charge states ranging from as high as +18 to as low as +5, depending on instrumentation used (Fig. 2.A

and 2.B). It is also demonstrated that calmodulin is capable of being ionized in the negative ion mode by MAI, producing charge states between –18 and –8 (Fig. S3). The experimentally determined MW's in both positive and negative detection modes are in excellent agreement with the theoretical value of 16706 Da. The added expense of lasers and safety requirements, as well as the higher background when using a laser, make MAI very attractive. For these reasons, the following studies were conducted in the Proteomics Core Facility at Wayne State University without the use of a laser.

The requirements for successful MAI are a proper matrix and temperature/pressure combination. Without instrumental modifications, the pressure of the experiment is usually fixed depending on whether the sample is introduced through the AP inlet or placed directly into vacuum. However, at least with the AP inlet, there is a range of temperatures that can be studied. An inlet temperature study on the Thermo Orbitrap Fusion shows the best ion abundance achieved for ubiquitin is at an inlet temperature of 100 °C (Fig. S4). Deconvolution programs developed for ESI can be employed to convert multiply charged ions into the molecular weight (MW) of the analyte, as shown for calmodulin (Fig. 2). Because of the ultra-high resolution of the Orbitrap Fusion used in this experiment (Fig. 2.C), the output of the protein mass measurement is with isotopic resolution when acquired with the resolution set to $m/\Delta m$ 450,000 (50% valley) at m/z 400. In addition, rapid acquisition of five proteins (ubiquitin, lysozyme, cytochrome c, calmodulin, and myoglobin) ranging in MW from 8.5 to 17 kDa is achieved within a minute (Fig. 3 and Fig. S5). The abundant highly charged protein ions are also readily accessible to be fragmented to allow sequence information to be obtained by using electron transfer dissociation (ETD), as is shown for ubiquitin charge state +12 obtained within ~20 s (Inset of Fig. S6). Because of the ultra-high resolution, the charge states of ETD generated fragments of high-mass ions are deciphered by the program automatically. The high resolving power is also useful for direct analysis of small molecules from complex mixtures. MAI-MS of the drug fexofenadine, in urine, and from blood wiped from skin surfaces is shown in Fig. S7 and Fig. 4, respectively. The background in the skin sample is related to synthetic polymers used in facial lotion. MAI produces few chemical background ions that are not sample related allowing the active ingredient of the drug to be readily detected.

An advantage of using MAI, especially relative to ESI, is that high salt content is well tolerated. A buffered, salty solution of the B-subunit of cholera toxin shows multiply protonated ions

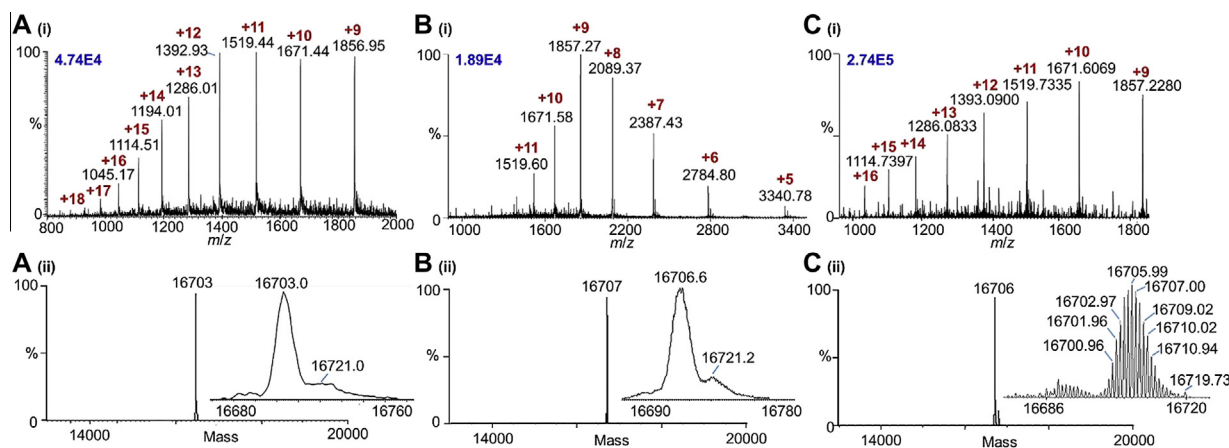


Fig. 2. Direct MAI-MS of 2.5 pmol protein calmodulin (Theoretical: 16706 Da/Experimental: Thermo LTQ Velos, 16703 Da; Waters SYNAPT G2, 16707 Da and Thermo Orbitrap Fusion, 16706 Da) extracted from *Escherichia coli* without cleanup procedures using 3-NBN as the matrix and different mass spectrometers at their optimal inlet temperatures: (A) Thermo LTQ Velos at 150 °C, (B) Waters SYNAPT G2 (mass resolution 1953 at m/z at 1857) at 40 °C and (C) Thermo Orbitrap Fusion (mass resolution of 40651 at m/z 1857) at 100 °C, the (i) mass spectra and (ii) deconvoluted spectra obtained using MagTran software including insets.

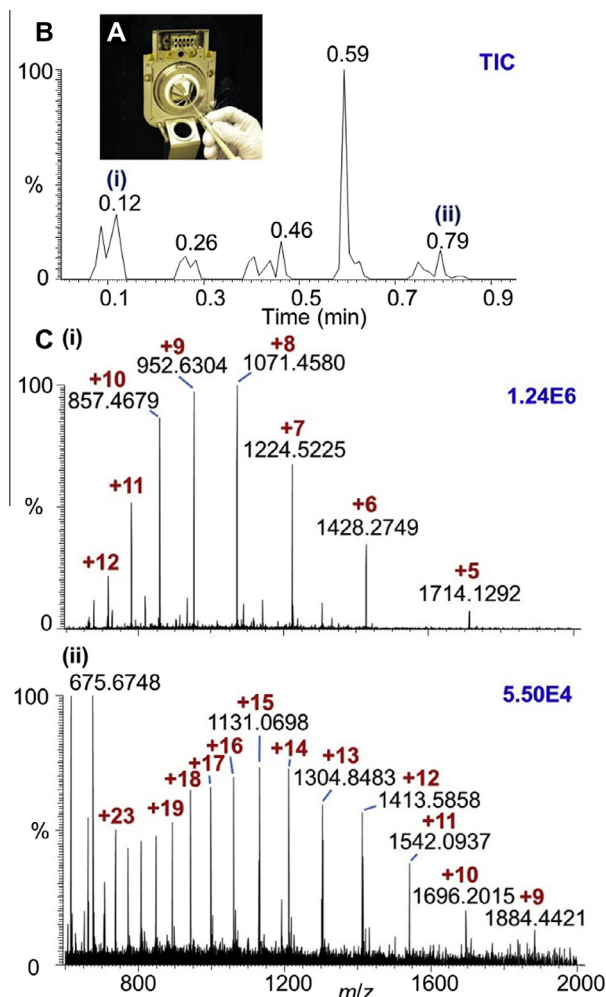


Fig. 3. Rapid high resolution MAI-MS of five different proteins: (A) photograph of the overridden ESI source of the Thermo Orbitrap Fusion operated at 100 °C inlet temperature, (B) TIC and (C) mass spectra of (i) ubiquitin (Theoretical: 8565 Da/Experimental: 8564 Da) and (ii) myoglobin (Theoretical: 16951 Da/Experimental: 16951 Da). Mass spectra of lysozyme (Theoretical: 14305 Da/Experimental: 14305 Da), cytochrome c (Theoretical: 12359 Da/Experimental: 12359 Da), and calmodulin (Theoretical: 16706 Da/Experimental: 16707 Da) can be found in Fig. S5. Matrix 3-NBN and an Orbitrap Fusion at 240,000 mass resolution was used. A range of 1 to 5 pmol of each sample was introduced per acquisition.

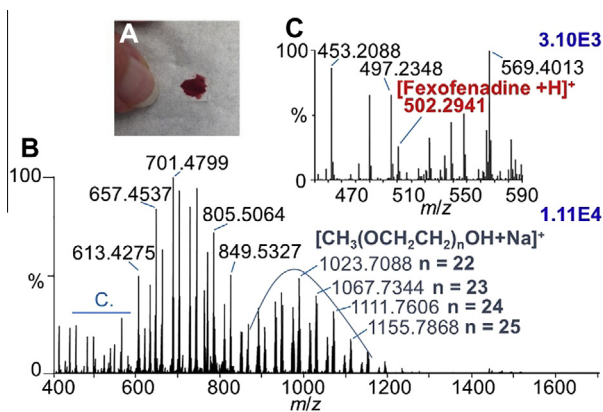


Fig. 4. MAI-MS analysis of a blood spot wiped from a minor facial scratch using a Kimwipe of a person using allergy medication containing Fexofenadine (MW 501.2879 Da). (A) Photograph, (B) full and (C) inset mass spectrum. A source temperature of 100 °C was used on a Thermo Orbitrap Fusion acquired at 450,000 mass resolution. All sample handling and analyses were performed by the person with the facial scratch.

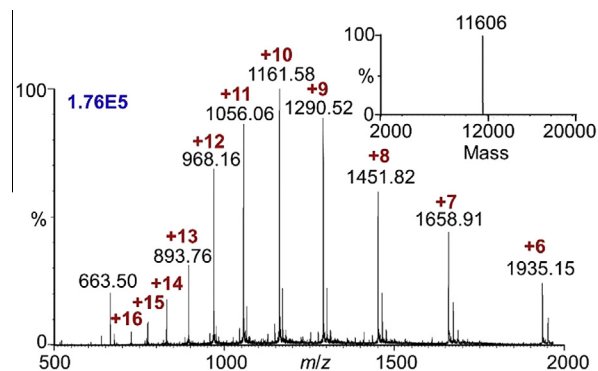


Fig. 5. MAI-MS of 1 pmol cholera toxin B-subunit in the presence of high amounts of NaCl. Inset: deconvolution of m/z to protein mass value obtained using MagTran software. The matrix 3-NBN and 30 °C inlet temperature were used on a Waters SYNAPT G2 mass spectrometer.

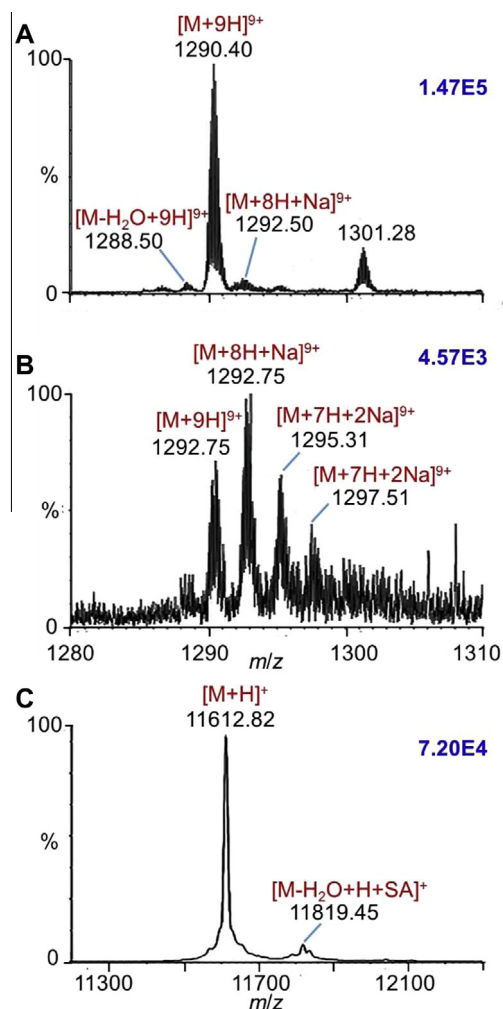


Fig. 6. Inset mass spectra of cholera toxin B subunit using different ionization methods: (A) MAI with 3-NBN with and inlet at 30 °C, (B) ESI, and (C) MALDI with sinapinic acid. MAI and ESI analyses were performed on a Waters SYNAPT G2, and MALDI-TOF analysis was performed on a Bruker Ultraflexstream mass spectrometer. MAI used a total of 0.5 pmol, ESI used 200 pmol of a solution at 1 pmol μL^{-1} , MALDI contained a sample spot with 5 pmol of analyte. Full mass spectra for these insets are included in Fig. S9.

up to the charge state of +16 with essentially no chemical background on a SYNAPT G2 mass spectrometer (Fig. 5); the deconvoluted mass is 11606 Da. Even in the presence of high amounts of NaCl and phosphate buffer, the amount of salt adduction observed is minimal. Cholera toxin B-subunit was readily detected (Fig. S8) at 25 fmol consumed without the need for purification or concentration prior to analysis. A direct comparison of the buffered solution of B-subunit of cholera toxin using different ionization methods (Fig. 6 and Fig. S9) show ion abundances readings for the most abundant charge state: $([M+9H]^{9+})$ of 1.47E5 for MAI, $([M+8H+Na]^{9+})$ of 4.57E3 for ESI, and $([M+H]^{+})$ of 7.20E4 for MALDI (please note that this data was acquired on two different mass spectrometers). While both MAI (Fig. 6.A) and ESI (Fig. 6.B) provide predominantly the protonated multiply charged ions, ESI has undesired interference of sodium adduction. The MALDI-TOF MS results (Fig. 6.C) show the typical broad protein signals and an ion associated with matrix adduction. These results indicate that MAI-MS is sensitive and robust towards salt and sample additives such as EDTA and sodium azide (NaN_3).

Another example of MAI for a natural protein is shown for the pertussis toxin, a bacterial toxin that exists as a 117 kDa hexameric protein complex made up of 6 subunits, denoted as S1 – S5 with the S4 subunit present in two copies (Fig. 7). Analyte solvent conditions influence the analysis using the same matrix, 3-NBN, and the same instrument acquisition conditions. From aqueous solution conditions one detects preferentially the S4 subunits of bacterial toxin with charge states of +12 to +6 (Fig. 7.A). The addition of methanol and acetic acid to the solvent showed a decrease in the ratio of protonated:sodium adducted signals compared to the preparation in water. Using the acidified methanol–water solvent

conditions, the MAI mass spectrum shows all of the subunits present (Fig. 7.B), with the S4 subunit being the most abundant ion. The observed m/z values for each subunit present in the mass spectrum are tabulated in Table S1. All of the species exist predominantly as multiply protonated ions, although some additional lower abundant peaks are observed due to the presence of sodium-adducted ions as well. Analysis of the B-subunit of pertussis toxin acquired using a MALDI-TOF mass spectrometer [23] resulted in singly charged ions of the S2 and S3 subunits (MW ca. 21.9 kDa), differing in mass by ca. 100 Da, which could not be resolved at such a high m/z because of the width of the protein signals associated with typical MALDI-TOF MS analysis [24,25]. With MAI however, multiple charging shifts the observed m/z towards the lower mass range for higher mass resolution measurements [3,4], allowing both the S2 and S3 subunits to be resolved by m/z using high resolution measurements (Fig. 7.B).

4. Conclusion

The results presented here show the potential of a new ionization method for analysis of nonvolatile biological compounds ionized directly from surfaces with (LSI) and without (MAI) the use of a laser. The new ionization methods lead to highly efficient ionization even for nonvolatile samples from ‘dirty’ conditions and, when performed on high-performance mass spectrometers, allow rapid deciphering of complex samples. Analyte solution conditions factor into the ion abundance of buffered and salty samples such as those of the membrane-associated bacterial toxins. The results presented here demonstrate that MAI and LSI have advantages relative to ESI and MALDI even though the experiments are carried out on instruments designed for these latter techniques. The success of LSI and MAI is related to the search for matrices with increasing volatility under vacuum conditions [5,22,26].

Acknowledgement

This work was made possible by the generous support from NSF CAREER 0955975, NSF CHE-1411376, DuPont Young Professor Award, Eli Lilly Young Investigator Award in Analytical Chemistry, Waters Center of Innovation Program, Thermo Scientific, and WSU Schaap Faculty Scholar (to ST) as well as grants to support the Core (and to PMS) P30 ES020957, P30 CA022453 and S10 OD010700. We would like to specifically thank Dr. Alexander Makarov and Jean-Jacques Donyach for their input with the Orbitrap Fusion (Thermo).

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jymeth.2016.01.019>.

References

- [1] A.J. Link, J. Eng, D.M. Schieltz, E. Carmack, G.J. Mize, D.R. Morris, B.M. Garvik, J. R. Yates, Direct analysis of protein complexes using mass spectrometry, *Nat. Biotechnol.* 17 (1999) 676–682.
- [2] J.P. Overington, B. Al-Lazikani, A.L. Hopkins, How many drug targets are there?, *Nat. Rev. Drug Discov.* 5 (2006) 993–996.
- [3] E.D. Inutan, S. Trimpin, Matrix assisted ionization vacuum (MAIV), a new ionization method for biological materials analysis using mass spectrometry, *Mol. Cell. Proteomics* 12 (2013) 792–796.
- [4] S. Trimpin, E.D. Inutan, Matrix assisted ionization in vacuum, a sensitive and widely applicable ionization method for mass spectrometry, *J. Am. Soc. Mass Spectrom.* 24 (2013) 722–732.
- [5] S. Trimpin, C.A. Lutowski, T.J. El-Baba, D.W. Woodall, C.D. Foley, C.D. Manly, B. Wang, C.-W. Liu, B.M. Harless, R. Kumar, L.F. Imperial, E.D. Inutan, Magic matrices for ionization in mass spectrometry, *Int. J. Mass Spectrom.* 377 (2015) 532–545.

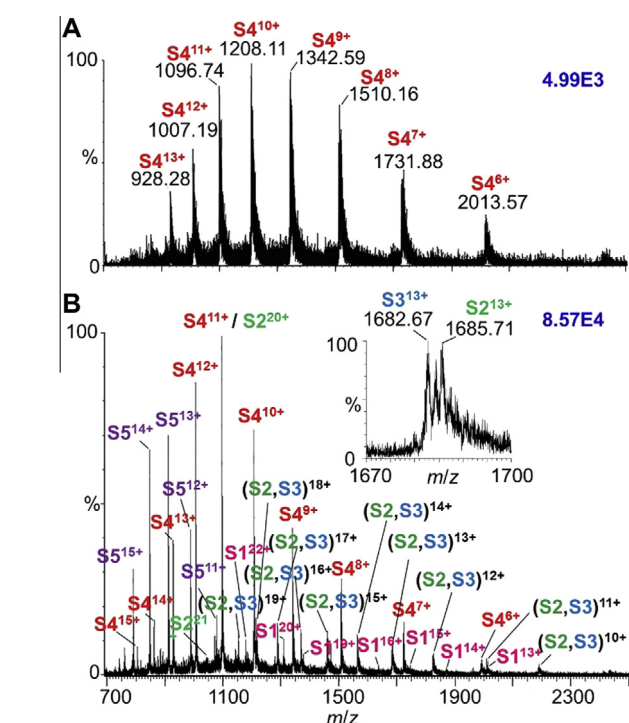


Fig. 7. MAI-MS of the hexameric pertussis toxin (mutant) consisting of five distinct subunits (S1 to S5) prepared in: **A.** 100% water and **B.** 50% MeOH with 1% acetic acid using a Waters SYNAPT G2 with an inlet at 30 °C; Inset: mass spectrum of subunits S2 and S3 with charge states +13. All subunits are present as indicated by the colored labels, S1 (pink, Theoretical: 26017 Da/Experimental: 26124 Da), S2 (green, Theoretical: 21839 Da/Experimental: 21907 Da), S3 (dark blue, Theoretical: 21751 Da/Experimental: 21864 Da), S4 (red, Theoretical: 12061 Da/Experimental: 12053 Da), and S5 (purple, Theoretical: 11747 Da/Experimental: 11838 Da). Table S1 summarizes the observed m/z values in (B) for all signals corresponding to each pertussis toxin subunit and used to determine each mass.

- [6] R.D. Smith, J.A. Loo, C.G. Edmonds, C.J. Barinaga, H.R. Udseth, New developments in biochemical mass spectrometry: electrospray ionization, *Anal. Chem.* 62 (1990) 882–899.
- [7] M. Sarbu, F. Zhu, J. Peter-Katalinic, D.E. Clemmer, A.D. Zamfir, Application of ion mobility tandem mass spectrometry to compositional and structural analysis of glycopeptides extracted from the urine of a patient diagnosed with Schindler disease, *Rapid Commun. Mass Spectrom.* 29 (2015) 1929–1937.
- [8] C.J. Tu, Q.H. Sheng, J. Li, D.J. Ma, X.M. Shen, X. Wang, Y. Shyr, Z.P. Yi, J. Qu, Optimization of search engines and postprocessing approaches to maximize peptide and protein identification for high-resolution mass data, *J. Proteome Res.* 14 (2015) 4662–4673.
- [9] M.J. Sweredoski, A. Moradian, M. Raedle, C. Franco, S. Hess, High resolution parallel reaction monitoring with electron transfer dissociation for middle-down proteomics, *Anal. Chem.* 87 (2015) 8360–8366.
- [10] S. Trimpin, E.D. Inutan, New ionization method for analysis on atmospheric pressure ionization mass spectrometers requiring only vacuum and matrix assistance, *Anal. Chem.* 85 (2013) 2005–2009.
- [11] S. Trimpin, E.D. Inutan, T.N. Herath, C.N. McEwen, Laserspray ionization – a new atmospheric pressure MALDI method for producing highly charged gas-phase ions of peptides and proteins directly from solid solutions, *Mol. Cell. Proteomics* 9 (2010) 362–367.
- [12] S. Trimpin, E.D. Inutan, T.N. Herath, C.N. McEwen, Matrix-assisted laser desorption/ionization mass spectrometry method for selectively producing either singly or multiply charged molecular ions, *Anal. Chem.* 82 (2010) 11–15.
- [13] S. Trimpin, "Magic" ionization mass spectrometry, *J. Am. Soc. Mass Spectrom.* 27 (2016) 4–21.
- [14] S. Trimpin, B. Wang, E.D. Inutan, J. Li, C.B. Lietz, V.S. Pagnotti, A.F. Harron, D. Sardelis, C.N. McEwen, A mechanism for ionization of nonvolatile compounds in mass spectrometry: considerations from MALDI and inlet ionization, *J. Am. Soc. Mass Spectrom.* 23 (2012) 1644–1660.
- [15] T.L. Constantopoulos, G.S. Jackson, C.G. Enke, Effects of salt concentration on analyte response using electrospray ionization mass spectrometry, *J. Am. Soc. Mass Spectrom.* 10 (1999) 625–634.
- [16] D.-C. Moon, J.A. Kelley, A simple desalting procedure for fast atom bombardment mass spectrometry, *Biomed. Environ. Mass Spectrom.* 17 (1988) 229–237.
- [17] M.G. Pluskal, Microscale sample preparation, *Nat. Biotechnol.* 18 (2000) 104–105.
- [18] S. Koster, E. Verpoorte, A decade of microfluidic analysis coupled with electrospray mass spectrometry: an overview, *Lab Chip* 7 (2007) 1394–1412.
- [19] W.E. Haskins, Z. Wang, C.J. Watson, R.R. Rostand, S.R. Witowski, D.H. Powell, R. T. Kennedy, Capillary LC–MS² at the attomole level for monitoring and discovering endogenous peptides in microdialysis samples collected in vivo, *Anal. Chem.* 73 (2001) 5005–5014.
- [20] D.W. Woodall, B. Wang, E.D. Inutan, S.B. Narayan, S. Trimpin, High-throughput characterization of small and large molecules using only a matrix and the vacuum of a mass spectrometer, *Anal. Chem.* 87 (2015) 4667–4674.
- [21] M. Pizza, A. Covacci, A. Bartoloni, M. Perugini, L. Nencioni, M.T. De Magistris, L. Villa, D. Nucci, R. Manetti, M. Bugnoli, F. Giovanoni, R. Olivieri, J.T. Barbieri, H. Sato, R. Rappuoli, Mutants of pertussis toxin suitable for vaccine development, *Science* 246 (1989) 497–500.
- [22] S. Trimpin, Y. Ren, B. Wang, C.B. Lietz, A.L. Richards, D.D. Marshall, E.D. Inutan, Extending the laserspray ionization concept to produce highly charged ions at high vacuum on a time-of-flight mass analyzer, *Anal. Chem.* 83 (2011) 5469–5475.
- [23] B. Liang, Y. Ju, J.R. Joubert, E.J. Kaleta, R. Lopez, I.W. Jones, H.K. Hall Jr, S.N. Ratnayaka, V.H. Wysocki, S.S. Saavedra, Label-free detection and identification of protein ligands captured by receptors in a polymerized planar lipid bilayer using MALDI-TOF MS, *Anal. Bioanal. Chem.* 407 (2015) 2777–2789.
- [24] S. Niu, W. Zhang, B.T. Chait, Direct comparison of infrared and ultraviolet wavelength matrix-assisted laser desorption/ionization mass spectrometry of proteins, *J. Am. Soc. Mass Spectrom.* 9 (1998) 1–7.
- [25] S. Trimpin, M.L. Deinzer, Solvent-free MALDI-MS for the analysis of a membrane protein via the mini ball mill approach: case study of bacteriorhodopsin, *Anal. Chem.* 79 (2007) 71–78.
- [26] C.N. McEwen, B.S. Larsen, S. Trimpin, Laserspray ionization on a commercial atmospheric pressure-MALDI mass spectrometer ion source: selecting singly or multiply charged ions, *Anal. Chem.* 82 (2010) 4998–5001.